

Course Plan: ENGR43301D Engineering Economics and Entrepreneurship

Semester: Fall 2026

Office Hours: Tuesdays, 11:00—12:00 PM

Instructor: Dr. Manju Sunil Varghese P.Eng

Week	Module	Topic	Activity	Engineering Tool
1	1	Fundamental Concepts – Chapters 1A and 2A -The role of engineering economic analysis; the decision making process; ethics; -Engineering Decision-Making for Current Costs -Engineering costs: fixed, variable, marginal, average, sunk, opportunity, recurring/non-recurring, incremental costs, life-cycle costs Cost Estimating; Estimating benefits; Cash flow diagrams		
2	1	Accounting and Engineering Economy -The Role of Accounting -The Balance Sheet -The Income Statement -Traditional Cost Accounting	Project (20 %) assigned- Sign up 4 group members	See deadlines later
3	2	Interest and Financial Mathematics – Chapters 3A & 4A -Simple Interest; Compound Interest; Repaying a debt; Equivalence -Single-payment Compound Interest Formulas: compound amount; present worth -Equivalence for repeated cash flow: uniform series compound amount; sinking fund; uniform series present worth; capital recovery; -Arithmetic Gradient -Gradient cash flow: gradient present worth; gradient uniform series		
4	3	Present Worth Analysis- Chapter 5A Present Worth Techniques: -Useful Lives Equal to the Analysis Period -Useful Different Lives from the Analysis Period -Infinite Analysis Period: Capitalized Cost	Individual Assignment 1 (10%)	
5	4	Annual Worth Analysis- Chapter 6A -Annual Cash Flow Calculations -Salvage Value -Annual Cash Flow Analysis: common multiples of alternatives; continuing requirement; infinite analysis period -Mortgages in Canada		
6	5	Benefit/Cost Ratio - Chapter 8A -Public Sector Investment Objectives -Selecting an Interest Rate: No time-value-of money, cost-of-capital, opportunity, recommended concepts -The Benefit/Cost Ratio	Case Study 1 (10 %) – work in groups of 2	

		-Incremental Benefit/Cost Analysis -Payback Period		
7	6	Rate of Return (ROR) Analysis- Chapter 7A -Minimum Attractive Rate of Return -Internal Rate of Return -Calculating Rate of Return: Plot of NWP vs. Interest Rate -Incremental Analysis -Sensitivity Analysis - Modified Internal Rate of Return	Project Report 1 (15%)of 20 %	EXCEL (Project)
		Break		
8	7	Entrepreneurial Opportunity – Chapter 1B Entrepreneurs -Entrepreneurial Opportunities -Entrepreneurship and Small Business -The Motivation, Drawbacks and Many Varieties of Entrepreneurship -The Changing Face of Entrepreneurship -Characteristics of Successful Entrepreneurs Family Enterprise – Chapter 3B -The Family Business: A Unique Institution -Family and Business Overlap -Family Business Momentum -Family Roles and Relationships -The Need for Good Management in the Family Firm -The Process of Leadership Succession		
9	8	Start-Up and Need for Competitive Advantage – Chapter 2B -The Start-Up: Creating a New Business -Finding Start-Up Ideas -Using Internal and External Analysis to Identify and Assess New Business Ideas -Strategies and Sustaining Competitive Advantage -Market Segmentation and Its Variables; Niche Market Franchising and Buyouts – Chapter 4B -Understanding the Franchise Concepts -The Pros and Cons of Franchising -Evaluating Franchise Opportunities -Selling a Franchise -Legal Issues in Franchising -Buying and Existing Business -Buying Assets -Quantitative Factors in Valuing the Business	Case Study 2 – Groups of 2 (10%)	
10	9	Developing an Effective Business – Chapter 5B -An Overview of the Business Plan -The Need for a Business Plan	Project Report 2	EXCEL (Project)

		-How Much Business Plan is Needed -Preparing a Business Plan -Advice and Resources of Writing Business Plan -Resources for Business Plan Preparation Intellectual Property (IP) and Patents; Definition of IP Copyright, Trade Secret; Patent Intellectual Property protection and filling	(15%) of 20%	
11	10	Sources of Financing– Chapter 14B -Firm Characteristics and Sources of Financing -Debt or Equity Financing: Potential Profitability, Risk -Sources of Early Financing -Banking Financing: Types of Loans -Understanding a Bank’s Perspectives/Formula Lending -Negotiating the Loan: IRR, Loan Maturity Date -Business Suppliers & Asset-Based Lending -Private Equity Investors -Government Programs & Agencies -Other Sources of Financing -Crowdfunding	Individual Assignment 2 (10 %)	
12		Presentation & Final Report due	Presentation (40% of 20 %) (3) & Project Report (7) due (30% of 20 %)	Report and EXCEL (Project)
13	NA	Exam Reviews	NA	
14		Final Exam (All Modules)	30%	

Assessment Criteria:

Cases Studies (2@10%)	20%	
Individual Assignments (2@10%)	20%	
Business Project	20%	
Final Exam	40%	(Modules 1 to 10)

Required Text:

Engineering Economic Analysis, Newnan, D., Jones, J., Whittaker, J., Eschenbach, T., and Lavelle, J., Oxford University Press, 4th Canadian Edition. ISBN 9780195430172, 2018

Small business management: Launching and Growing New Ventures, Longenecker, J. G., Donlevy, L. B., Champion, T., Petty, J. W., Leslie, E. P., & Hoy, F. Cengage, 6th Canadian Edition, ISBN 978-0-17-653221-5, 2021

Note 1: The project is on group basis, requiring a portfolio to be prepared by each group, but Project Plan must show each student allocation; there will be presentation.

Note 2: It is the student's responsibility to be familiar with the Missed Evaluations Process on SLATE. These are binding.